

Yuancheng (Remo) Shen

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EDUCATION

- **New York University** New York, United States
Ph.D. - Computer Science Sep. 2024 - present
Advisor: Prof. Robert Krueger
- **Shandong University** Shandong, China
Master - Computer Science Sep. 2021 - Jun. 2024
Advisor: Prof. Yunhai Wang
- **Jiangsu University of Science and Technology** Jiangsu, China
Bachelor - Computer Science and Technology Sep. 2017 - Jun. 2021
GPA: 3.85; Ranking: 2/102

PUBLICATIONS

- [P1] **Y. Shen**, C. Kalsi, P. Motarwar, D. Annuar, J. Stimes, S. Sterling, S. Qiu, I. Illa-Bochaca, I. Osman, D. Fenyo, G. J. Baker, E. Hernando, K. Ruggles, A. Tsirigos, and R. Krueger. CellGalaxy: Multiscale Exploration of Biomedical Image Embeddings. In *IEEE Transactions on Visualization and Computer Graphics*. (Under Review).
- [P2] Yunge Wen, Yaluo Wang, **Yuancheng Shen**, Robert krueger, Paul Liang. PaintCopilot: Modeling Painting as Autonomous Artistic Continuation. (Under Review).
- [P3] C. Kalsi, H. Fathollahian, L. Reichman, S. Gaupp, **Y. Shen**, A. Shabanova, A. Junquera, A. Szabó, A. Rodríguez, T. Vallius, M. Rogava, G. E. Marai, M. Krone, S. Boorboor, and R. Krueger. BioSET: Interactive Visual Analysis of Biomarker Co-localization in Large Multi-Volume Tissue Data. In *IEEE Transactions on Visualization and Computer Graphics*. (Under Review).
- [P4] Yunge Wen, **Yuancheng Shen**, Robert krueger, Paul Liang. PaintCopilot: Modeling Painting as Autonomous Artistic Continuation. In *NeurIPS Creative AI, 2026*. (Under Review).
- [P5] C. Kalsi, **Y. Shen**, S. Gaupp, L. Reichmann, M. Rogava, M. Krone, S. Boorboor, and R. Krueger. BioSET - Biomarker-based Spatial co-Expression analysis in Tumor environments. In *Bio+MedVis Challenge @ IEEE VIS, 2025 (Award of Excellence)*.
- [P6] **Yuancheng Shen**, Yue Zhao, Yunhai Wang, Tong Ge, Haoyan Shi and Bongshin Lee. Authoring Data-Driven Chart Animations through Direct Manipulation. In *IEEE Transactions on Visualization and Computer Graphics, 2024*. (DOI: 10.1109/TVCG.2024.3491504).
- [P7] Tao Dai, Qi Wang, **Yuancheng Shen**, Shang Gao. SwinVision: detecting small objects in low-light environments. In *IEEE Access, 2025* (DOI: 10.1109/ACCESS.2025.3548151).

RESEARCH EXPERIENCE

- **Visual Analytics for Spatial and Cellular Imaging** New York University
Advisor: Prof. Robert Krueger Sep. 2024 – Present
 - **Interactive and LLM-assisted visual analytics for biomedical imaging:** Investigate how large-scale cellular imaging datasets can be explored interactively across multiple biological modalities, including multiplex tissue microscopy, spatial transcriptomics, and live-cell imaging. Our work studies multiscale visual representations that connect image embeddings with spatial tissue organization and temporal cellular dynamics, while integrating AI/LLM-assisted interpretation to support discovery of emerging biological patterns. Conducted in collaboration with biomedical researchers at NYU Langone Health, Harvard Medical School, Northwell Health, and Oregon Health & Science University.
 - **Deep learning-based analysis of 3D cellular imaging:** Study how volumetric microscopy data can be transformed into quantitative cellular representations for downstream biological analysis. Our work investigates learning-based analysis pipelines integrating 3D cell segmentation, biomarker quantification, and spatial feature extraction, enabling systematic characterization of cellular microenvironments in large multiplex tissue datasets.
 - **Representation learning for cellular morphology and microenvironments:** Investigate representation learning methods for modeling 3D cellular morphology and spatial tissue context. Our work explores vision transformer-based architectures to capture long-range structural relationships in volumetric cellular data, enabling tasks such as cell registration, similarity search, and cross-sample comparison of cellular organization.
- **Human-Centered Authoring of Data-Driven Chart Animations** Shandong University
Advisors: Yunhai Wang and Bongshin Lee Oct. 2022 – Jun. 2024
 - **Domain-specific languages for chart animation specification:** Study how expressive chart animations can be specified through high-level declarative languages rather than low-level keyframe manipulation. Our work contributes a domain-specific language for data-driven animations that models animation timing, mark grouping, and visual effects, enabling concise specification of complex animated visualizations.

- **Interactive systems for animation authoring:** Develop interactive authoring systems that enable users to create chart animations through direct manipulation and visual programming. Built upon the animation specification language, these systems support expressive animation design while lowering the barrier for non-programmers to create data-driven animated visualizations.
- **Learning-Based Visual Detection in Challenging Environments** Shandong University
Advisor: Shang Gao *Jan. 2024 – Sep. 2024*
 - **Transformer-based small object detection:** Investigate deep learning approaches for detecting small objects in challenging low-light environments, where weak visual signals and noise significantly degrade object recognition. We propose a transformer-based detection framework that enhances global contextual modeling and multi-scale feature representation, integrating low-light enhancement and attention mechanisms to improve detection accuracy while maintaining computational efficiency.

SELECTED HONORS AND AWARDS

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- NYU School of Engineering (SOE) PhD Fellowship, New York University *Sep. 2024*
 - Outstanding Thesis Award, Shandong University *Jun. 2024*
 - Outstanding Graduates Award, Shandong University *Jun. 2024*
 - Postgraduate Excellent Student Award Fund, Shandong University *Oct. 2021*
 - Outstanding Thesis Award, Jiangsu University of Science and Technology *Jun. 2021*
 - Outstanding Graduates Award, Jiangsu University of Science and Technology *Jun. 2021*
 - 1st Prize Scholarship, Jiangsu University of Science and Technology *Oct. 2019*
 - 1st Prize in Higher Mathematics Competition, Jiangsu *Aug. 2018*

REVIEWS AND COMMUNITY WORK

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- **Reviewer:** IEEE Vis; ACM CHI; Expert Systems With Applications; PeerJ Computer Science

ACADEMIC ENGAGEMENTS

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- **International Conference on Geometric Modeling and Processing** Qingdao, China
Participated in the event, received experts and scholars, and volunteered for other conference services *Jun. 2024*
 - **The Geometric Design and Computing Conference** Qingdao, China
Participated in the event, received experts and scholars, and volunteered for other conference services *Aug. 2022*
 - **The China Visualization and Visual Analytics Conference** Xining, China
Participated in the event *Jul. 2022*

SKILLS

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- **Programming:** Python, TypeScript, JavaScript, C++, SQL, R
 - **Machine Learning:** PyTorch, NumPy, Pandas
 - **Visualization & Tools:** WebGL, VTK.js, D3.js, Git, Linux, LaTeX, Adobe Illustrator